

From Comparing Groups to Modeling Relationships

Module 4, Part 1 (November 13)

QBA 5330

Where We're Headed Today

- **Part 1:** From comparing groups to modeling relationships
- **Part 2:** Diagnostics: can this model be trusted?
- **Part 3:** Logistic regression and the Module 4 deliverable

Module 3 vs. Today

- Module 3: comparing groups (means, proportions), ANOVA, causality
- Today: *modeling relationships*, predicting one variable from another

The Analytics Decision Cycle, Revisited

Frame the decision → obtain data → summarize evidence
→ quantify uncertainty → **build a model** → recommend
an action → communicate the result

- Everything through Module 3 sat in steps 1–4
- Today: step 5

A Familiar Dataset, One Predictor

$$\widehat{\text{sales}} = 10.50 + 0.0527(\text{tv})$$

```
1 library(tidyverse)
2 ads <- read_csv("data/advertising.csv")
3 model_tv <- lm(sales ~ tv, data = ads)
4 summary(model_tv)$r.squared
```

```
[1] 0.713791
```

What's Missing?

- The company also spends on **radio** and **online** advertising
- A model using only **tv** might be silently absorbing their effects too

Adding Predictors, Live

```
1 model_full <- lm(sales ~ tv + radio + online, data = ads)
2 summary(model_full)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.41940578	0.392732155	16.345506	1.867068e-38
tv	0.05123458	0.001594967	32.122663	4.976503e-80
radio	0.13268947	0.009268839	14.315652	2.744423e-32
online	0.03009225	0.005650494	5.325596	2.744114e-07

Watch R-Squared Climb

Model	R^2	Adjusted R^2
tv only	0.714	0.712
tv + radio	0.856	0.854
tv + radio + online	0.874	0.872

- Adjusted R^2 rises **right alongside** raw $R^2 \rightarrow$ real signal, not padding

The tv Coefficient Actually Moved

- Alone: $b_{tv} = 0.0527$
- With radio and online in the model: $b_{tv} = 0.0512$
- “Holding other predictors constant” is not just a phrase, it changes the number.

Quick Recap: Categorical Predictors

- A dummy coefficient is a comparison to the reference category
- Changing the reference category changes what's *reported*, not what's *predicted*

Activity: Interpret This Output (20 minutes)

A snack company surveys 200 households on income, region, and daily servings of fruits/vegetables.

In groups: what does this model actually tell the company?

The Output

household_income	-1.277e-06	p = 0.845
regionNorth	0.0342	p = 0.879
regionSouth	-0.3555	p = 0.102
regionWest	-0.1654	p = 0.449

Adjusted R-squared: 0.0001

Debrief

- What *is* the honest conclusion here?
- Is “we found nothing” still a useful answer for the company?

Before the Break

- We've been trusting every t-test and confidence interval at face value
- **Hour:** is that trust actually earned?